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Scalable spatial transcriptomics through computational array reconstruction

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Spatial transcriptomics enables gene expression mapping within tissues but is often limited by imaging constraints. We present an imaging-free approach that reconstructs spatial barcode locations using molecular diffusion and dimensionality reduction. Validated against ground truth imaging, our method achieves high fidelity and scales to centimeter-sized tissues. This approach enhances spatial transcriptomics' accessibility and throughput, enabling large-scale studies without specialized imaging equipment.

Tissue functions arise from the coordinated activities of cells. Interactions at multiple scales—from the level of molecules to communication between cells, to signaling across tissues—are all necessary for this coordinated function. Spatial transcriptomic technologies, which enable spatial localization of gene expression profiles within tissue contexts, represent a powerful set of tools to study tissue function and cell-cell interactions[1–5]. Currently, imaging is crucial in spatial transcriptomics for locating RNA molecules within tissues or indexing capture arrays. However, imaging requires specialized techniques and equipment, limiting throughput, adaptability and detectable area size[6]. Alternatively, arrays may be deterministically printed through lithography or physical methods, but such methods require complex equipment and high upfront costs[3,7]. An imaging-free spatial transcriptomic technique, which ideally can be performed without complex equipment, can enhance the throughput and accessibility of experiments and enable larger-scale detection for comprehensive studies of tissues.

In theory, spatial information can be inferred independently of imaging. Molecular proximity measurements capture interactions (for example, HiC[8]), whereas pairwise distance measurements can directly infer spatial locations. For example, a phone's location is determined by its distances to satellites. Similarly, the US geographical layout can be mathematically reconstructed from city-to-city distances[9]. Distance can also extend beyond Euclidean space—human genetic variations often correlate with geography, as seen in European genetic maps resembling actual geography[10]. At the molecular level, diffusion patterns, highlighting neighboring information, enable reconstruction of molecular locations[11–14]. For example, DNA microscopy infers spatial information from native molecular interactions without traditional imaging techniques[13]. Although these methods expand spatial measurement techniques, most remain theoretical[15,16] or limited to simplified experimental systems[13].

Here, we present an imaging-free spatial transcriptomics method that computationally reconstructs array barcode locations used in spatial transcriptomics measurements with high resolution and fidelity. We implemented the imaging-free approach on two-dimensional (2D) barcode arrays, along with ground truth imaging for error estimation, and used dimensionality reduction for spatial reconstruction. We demonstrated that this imaging-free strategy can be integrated with existing barcode-array-based spatial transcriptomics methods without perturbing spatial structures, enabling high-throughput barcode array generation and accessibility for labs without specialized imaging. Furthermore, we applied this technique to a tissue sample on the centimeter scale, showing its potential for large-scale spatial transcriptomics.

Results

Spatial reconstruction through dimensionality reduction

We started by asking if we could computationally reconstruct spatial locations in spatial transcriptomics using diffusion-based proximity data. Because Slide-seq, a high-resolution spatial transcriptomic approach, uses arrays of barcoded beads for spatial capture, we reasoned that we could simulate diffusional interactions between beads on a 2D array (Fig. 1a). To visualize the reconstruction effect, we uniformly sampled capture and diffusible beads from a circular area with the color pattern of the letter 'H' (Fig. 1b and Extended Data Fig. 1a). The diffusion of barcodes from each diffusible bead was simulated to follow a

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Gaussian distribution (Extended Data Fig. 1b and Methods). Because of this diffusion, capture beads exhibit proximity-dependent capture of barcodes from diffusible beads, generating a neighborhood matrix between bead barcodes (Fig. 1a). Specifically, capture beads registered higher count values for barcodes from diffusible beads that were closer. Although each capture barcode can be characterized by count values on its associated diffusible barcodes (a high-dimensional vector), physically adjacent capture beads capture barcodes from similar diffusible beads. We reasoned that dimensionality reduction algorithms may be able to reconstruct the latent 2D representation of physical space from this high-dimensional interaction matrix. In particular, we performed Uniform Manifold Approximation and Projection[17] (UMAP), a nonlinear dimensionality reduction method, which reduces the high-dimensional capture bead barcode by diffusible bead barcode data into a 2D embedding space while preserving the similarity of capture bead barcodes in high-dimensional space (Extended Data Fig. 1). Although all parameters clustered by proximity, we found a range of parameters (Supplementary Fig. 1) wherein locations of capture beads in this 2D embedding are highly similar to their physical locations, indicating UMAP learns the intrinsic 2D manifold embedded within the high-dimensional data (Fig. 1b and Extended Data Fig. 1c,d). The accuracy of reconstruction was quantitatively assessed by comparing the reconstructed locations of the capture beads to their known physical positions, following a rigid transformation for alignment. The median error in simulation was found to be equivalent to 1.6% of the array's diameter (Extended Data Fig. 1e–g and Supplementary Fig. 1b). Other dimensionality reduction methods (linear and nonlinear) were assessed using identical simulation data (Supplementary Fig. 2). Although all methods could discern the spatial structure of bead arrangements, they incurred larger absolute errors when compared to the results of UMAP.

To assess the practical applicability of this computational reconstruction method, we explored the sensitivity to experimental parameters in simulation. We analyzed reconstruction errors with different ratios of diffusible to capture bead numbers. The simulation results displayed median errors less than 2% of the array size for ratios ranging from 1:5 to 5:1, suggesting robustness to relative ratios of the two bead populations (Fig. 1c). We also simulated the effects of changing the number of captured unique molecule identifiers (UMIs) per capture bead and diffusion distance (σ) (Fig. 1c). We were encouraged that a wide range of parameters demonstrated feasibility for computational reconstruction, for example diffusion distance (σ) between 2% to 6% of the array size, and a minimum of 40 UMIs per bead.

Spatial transcriptomics through computational reconstruction

We next sought to implement our reconstruction strategy experimentally by generating a Slide-seq array with a mixture of diffusible and capture beads. To enable diffusion-based reconstruction, we designed a mixed array incorporating barcoded poly(dT) beads for capturing mRNA with barcoded poly(dA) diffusible beads at a ratio of 3 to 1 (Fig. 1a,d). To obtain spatial coordinates of beads by reconstruction, the oligonucleotide barcodes on poly(dA) beads (diffusible beads) were cleaved with UV light, enabling diffusion to nearby poly(dT) beads (capture beads) (Fig. 1d). To validate the reconstruction data, we generated ground truth positions of the same array before reconstruction using in situ sequencing[18]. With ground truth locations, the distribution of distances between capture beads and captured diffusible beads followed a heavy tailed distribution, with the full width at half-maximum (FWHM) around 123.1 µm (Fig. 1e).

We next sought to reconstruct the relative spatial locations using diffusion information. We applied UMAP on the high-dimensional barcode-interaction data to reconstruct the relative locations of capture beads in a 2D embedding space, without any spatial information input. In addition to the two main UMAP parameters (n_neighbors, min_dist) we tuned in simulation, we also found increasing the number of epochs, using cosine metric for computing high-dimensional distance, and applying log1p transformation of diffusion matrix improved the reconstruction accuracy (Supplementary Figs. 3 and 4). Comparing to ground truth locations, reconstructed locations recovered the global arrangement of capture beads (Supplementary Fig. 3a). We noticed a few beads (20 out of ~16,000) were positioned notably away (>200 µm) from ground truth positions, most of which were attributable to in situ sequencing errors (Supplementary Fig. 5) or barcode collisions.

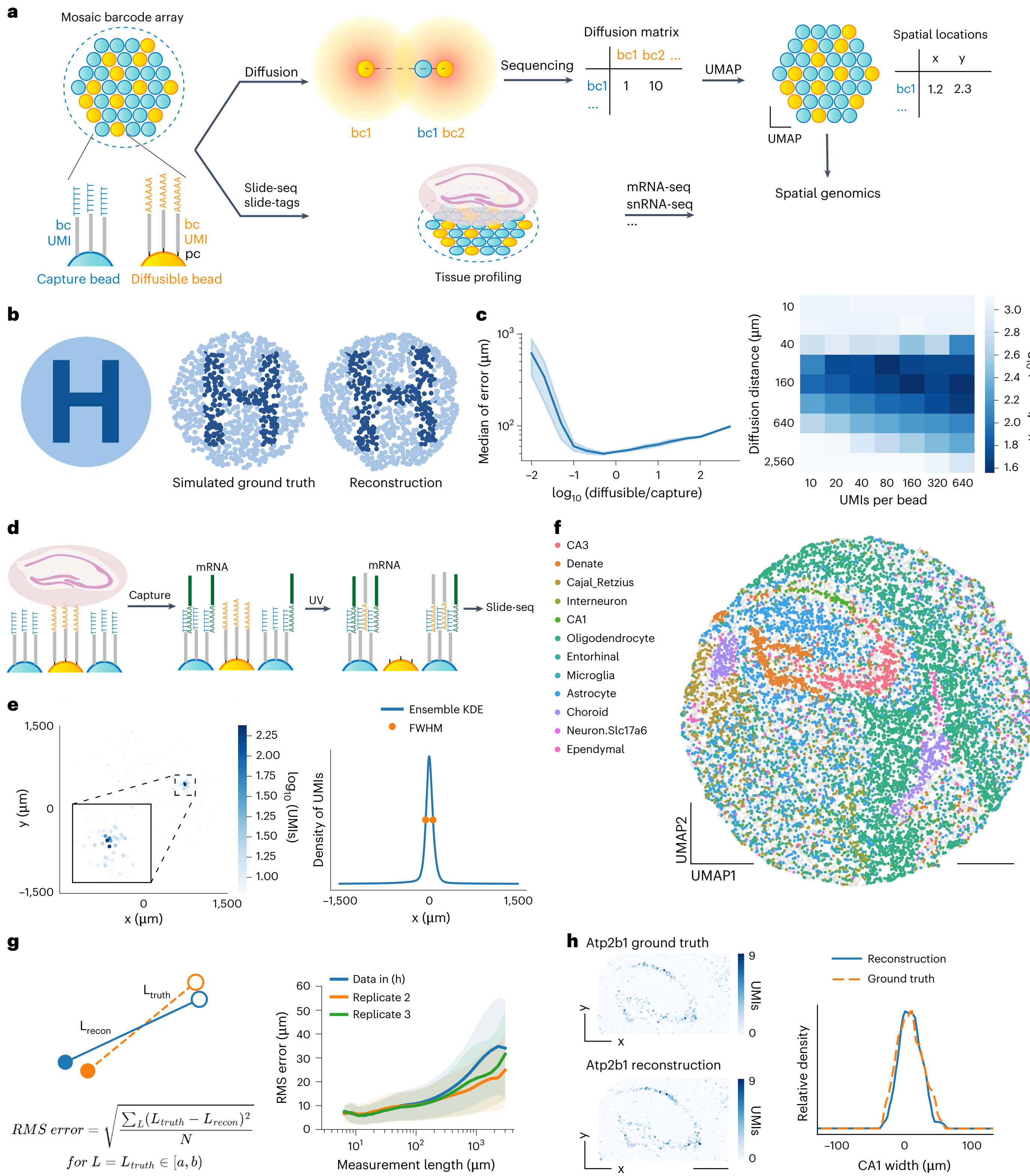
Given the reconstructed bead locations, we next examined spatial transcriptomics data captured by the array. We performed Slide-seqV2 using the same reconstructed array with the capture beads (poly(dT)) on a mouse hippocampal transcript with no modifications to the Slide-seq protocol. Individual bead profiles were clustered and assigned with cell types by robust cell-type decomposition[19] (RCTD). Spatial representation of cell types with reconstructed capture bead location demonstrated the known structures of the hippocampus (Fig. 1f and Extended Data Fig. 2d), evident in the cell-type distributions and marker gene plots, which are virtually indistinguishable between reconstruction and ground truth data (Supplementary Figs. 6 and 7).

We next sought to quantify the accuracy of the reconstruction process with three strategies: examining the absolute error, relative error, and histological structure preservation in comparison to ground truth. First, to assess absolute error, we calculated each capture bead's absolute displacement via a rigid registration between reconstruction result and ground truth. The median value of displacement lengths was 25.9 µm (Extended Data Fig. 2a–c). The absolute error can be affected by the registration process; this suggests that a more appropriate statistic should be the error in distance measurements between reconstruction and ground truth (Fig. 1g, Extended Data Fig. 2e). We quantified the root mean-square (RMS) error of length measurements across all pairwise length measurements as a function of measurement length.

Fig. 1 | Imaging-free spatial transcriptomics through computational reconstruction. a, Schematic of imaging-free reconstruction for spatial transcriptomics. A mosaic barcode array contains capture beads (poly(dT)) and diffusible beads (poly(dA)), which are DNA-barcoded with a bead-specific spatial barcode (bc) and a UMI. Photocleaved diffusible barcodes diffuse to capture beads, forming a diffusion matrix. UMAP reconstructs bead locations, enabling Slide-seq/Slide-tags spatial transcriptomics. pc, photocleavable linker. **b**, Simulated reconstruction. (Left) Original image of letter 'H'. (Middle) Ground truth. (Right) Reconstructed bead locations. Beads are colored the same. **c**, (Left) Simulated reconstruction error as a function of the ratio of diffusible to capture beads. The error represents the median displacement distance of capture beads, with the whole array's diameter as 3 mm. The shaded area represents one standard deviation ($n = 5$ repeated simulations). (Right) Error as a function of diffusion distance and number of molecules carrying diffusion information (UMIs) per bead. **d**, Schematic of reconstruction for Slide-seq. **e**, (Left) Diffusion pattern of a capture bead barcode on its associated diffusible bead barcodes, colored by the number of unique joint UMIs. Inset detailing the diffusion center. (Right) Diffusion distribution in x-axis direction. Orange dots represent the half-maximum, corresponding to the full width at half-maximum (FWHM) measured as 123.1 µm. KDE, kernel density estimation. **f**, Slide-seq on a mouse hippocampus with reconstruction, colored by decomposed cell types. Locations were scaled to the original 3 mm diameter. **g**, (Left) Schematic of RMS error at different measurement lengths. Error is defined as the difference between the measured distance in ground truth (orange dashed line) and reconstruction (blue line). Errors are averaged by root mean-square (RMS). (Right) RMS error of different measurement lengths. Data shown in (**f**) (blue) and two biological replicates (orange and green) are presented. Solid lines represent mean values, and shaded areas represent one standard deviation. **h**, (Left) Spatial expression of Atp2b1, a CA1 layer marker, in ground truth and reconstruction. (Right) Representative plots of CA1 layer width by profiling the expression intensity of Atp2b1 along a perpendicular line in ground truth (orange) and reconstruction (blue). Scale bars: 500 µm.

RMS error was close to 10 μm (the bead size) at local scale measurements (~ 100 μm) as nearby beads were usually displaced in the same direction, and plateaus at ~25 μm >1000 micron (representing <2.5% error in measurement lengths) (Extended Data Fig. 2e).

To assess the effect of reconstruction error on histological structure, we measured the width of the CA1 layer in the hippocampus. The widths, characterized by CA1 marker Atp2b1 expression, were similar in ground truth and reconstruction (FWHM is 49.5 μm in ground truth and 43.7 μm in reconstruction) (Fig. 1h). We also performed neighborhood enrichment analysis (Methods) between all pairs of cell types and found the results highly similar between reconstruction and ground truth (with Pearson correlation coefficient = 0.997) (Extended Data Fig. 2g). Lastly, we evaluated the reconstruction error across three biological replicates in the hippocampus and found that results were largely concordant across replicates, thus demonstrating the robustness of the reconstruction procedure (Fig. 1g, Extended Data Fig. 2f).



$$RMS\ error = \sqrt{\frac{\sum_L (L_{truth} - L_{recon})^2}{N}}$$
$$for\ L = L_{truth} \in [a, b)$$

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These data demonstrate that the reconstruction error is relatively small (~ 2.5% - 10% of measurement lengths across length scales), with a subtle effect on local spatial transcriptomics analysis.

Spatial reconstruction at single-nucleus resolution

As the computational reconstruction strategy should be adaptable to all array-based spatial technologies, we next sought to apply reconstruction with Slide-tags, a recently developed single-nucleus spatial technology[20].

In Slide-tags, nucleic acid spatial barcodes are photocleaved from barcoded arrays, associated with nuclei, followed by single-nucleus RNA sequencing with single-cell indexing. To apply reconstruction to Slide-tags, we generated arrays with mixtures of photocleavable poly(dA) (diffusible) and non-cleavable poly(dT) (capture) beads at the ratio of 3 to 1. In the Slide-tags experiment, the diffusible beads (poly(dA)) are photocleaved, diffused and captured by capture beads while tagging nuclei at the same time (Fig. 2a).

We performed reconstruction Slide-tags on a coronal mouse hippocampal section. Again, in addition to computational reconstruction, we performed in situ sequencing to generate ground truth spatial positions. We generated high-quality single-nucleus spatial data (2,091 genes per cell), which show high-quality clustering of major cell types in the hippocampus (Fig. 2b and Supplementary Fig. 8). We next reconstructed the bead locations with diffusion information and positioned each nucleus based on bead barcodes that it is tagged with. Following reconstruction and spatial placement, the spatial representation of different cell types captured the architecture of the hippocampus (Fig. 2c).

We next evaluated the accuracy of the reconstruction of Slide-tags. We calculated the absolute positioning errors of each bead and each nucleus by comparing to ground truth locations from in situ sequencing, after applying a rigid transformation (Extended Data Fig. 3). We found that the median value of bead positioning error (25.4 µm) and nuclei positioning error (27.2 µm) are similar (Extended Data Fig. 3). To remove the error introduced by registration, we also calculated the RMS error of measurement lengths, which was smaller than 25 µm at local measurement scales (<500 µm) and plateaus at ~30 µm >1,000 microns (representing <3% error in measurement lengths) (Fig. 2d and Extended Data Fig. 3g). To assess the effect of reconstruction error on biological structure, we compared the spatial representation of each clustered cell type in reconstruction and ground truth, with no detectable differences (Supplementary Fig. 9). Lastly, Slide-seq and Slide-tags showed similar reconstruction error, likely due to the shared reconstruction protocol.

Reconstruction enables scalable spatial transcriptomics

Reconstruction is purely performed through molecular biology reactions, and, thus, is not limited by imaging throughput. To demonstrate the scalability of reconstruction for spatial transcriptomics, we performed reconstruction on a 1.2-cm P1 mouse cranial section with Slide-seq (Supplementary Video 1). We spatially profiled transcriptomics across different tissue types, including brain, muscle and the upper respiratory system, with a single section (Fig. 2e). When compared with hematoxylin and eosin staining of an adjacent section, reconstruction successfully identified the compartmentalization of different tissues and elucidated fine structural details (Fig. 2e). We assigned decomposed cell types to each bead with RCTD (Fig. 2f, Supplementary Fig. 10c and Supplementary Fig. 11). To assess the fidelity of the cell-type assignments, we plotted the spatial distribution of certain cell types that exhibit unique spatial localization: adipocytes around the anterior cervical region, neuronal cells in the central nervous systems (CNS), peripheral nervous systems (PNS) and the olfactory sensory region (Fig. 2g). The locations of these cell types are highly correlated with the spatial expression pattern of their maker genes (Fig. 2h). For instance, the locations of fibroblasts and osteoblasts correspond with the distribution of type I collagen, which is abundantly present in tendons and bones. Subclustering of neuronal assigned beads revealed distinctions between CNS, PNS and olfactory neurons, which were highly correlated with the expression pattern of their respective marker genes. We found cortical neurons with this higher resolution of clustering (Fig. 2i,j and Supplementary Fig. 10d–f).

As our dataset covers a wide range of tissue types, we sought out to identify cell-type-specific differential gene expression across the entire section. We first focused on the olfactory epithelium for its well-organized structures and multilayered cell compositions. We profiled the spatial expression of previously identified genes and found the result consistent with in situ hybridization[21]: carbonyl reductase 2 (*Cbr2*), a sustentacular cell maker, has higher expression in olfactory epithelium's outer layer; regenerating islet-derived protein 3 gamma (*Reg3g*), a respiratory epithelium marker, has decreased expression at the boundary between respiratory epithelium and olfactory epithelium; and growth associated protein 43 (*Gap43*), which marks immature olfactory sensory neurons (OSNs), is only expressed in the inner olfactory epithelium layer where immature OSNs are located (Supplementary Fig. 12a). Furthermore, by analyzing genes enriched in the olfactory epithelium, we identified additional spatially variable genes in the olfactory epithelium (Supplementary Fig. 12b). The spatial profiling of spatially differential expression genes in olfactory epithelium further displayed the high fidelity of our reconstruction method at fine structural scales.

We also performed nonparametric cell-type-specific inference of differential expression[22] (C-SIDE) of epithelial specific genes to identify genes with spatially variable expression. We also calculated Moran's I statistics, a spatial autocorrelation measurement, of genes with high C-SIDE Z-scores. The results from both analyses indicate a nonrandom pattern in the variable expression of these genes (Fig. 2k). Spatial representation of selected genes displayed their distinct expression patterns across various regions (Fig. 2l). For example, BPI fold containing family A member 2 (*Bpifa2*) is prominently expressed in the parotid gland epithelial cells. In addition, we analyzed spatial differential expression in

Fig. 2 | Reconstruction enables diverse spatial transcriptomics measurements at scale. a, Schematic of reconstruction for Slide-tags. **b**, UMAP embedding of snRNA-seq profiles from a coronal mouse hippocampus section, colored by cell-type annotations shown in **c**. **c**, Spatial location of nuclei mapped based on reconstructed locations of capture beads, colored by cell-type annotations. Locations were scaled to the original array size with a diameter of 3 mm. **d**, RMS length measurement error of nuclei in reconstruction versus ground truth, where solid line shows mean value, and shaded area is one standard deviation. **e**, (Left) Spatial transcriptomics of a P1 mouse head section from a single experiment. Beads are colored by decomposed cell-type annotations from RCTD. (Right) Hematoxylin and eosin staining of an adjacent slice from the same sample. Dashed line indicates the 1.2 cm bead array used for this experiment. **f**, UMAP representing gene expression of the mouse sample captured by beads, colored by decomposed cell types from RCTD. **g**, Spatial distribution of labeled cell types, colored the same as cell-type annotations in **f**. Gray beads represent other cell types, plotted for contrasting. Neuronal cells include olfactory sensory neurons, intermediate neuronal progenitors, CNS neurons, and neural crest PNS neurons from clusters in **e**. Chond, chondrocytes; Fibro, fibroblasts. **h**, Marker gene spatial expression of cell types shown in **h**. Colored by normalized and log1p-transformed gene expression level. **i**, Subclustring of neuronal cells, colored by subtype annotations. Gray beads are other subclusters of neuronal cells. **j**, Maker gene of each neuronal subtype shown in **i**, colored with corresponding color gradients based on normalized and log1p-transformed gene expression level. All beads of neuronal cell type are plotted. **k**, Within beads categorized under the epithelial type, top 20 spatially differential expression genes ranked by nonparametric C-SIDE were plotted, with Moran's I statistics calculated. Higher scores on both metrics signify more spatially variable expression. **l**, Spatially differential expression of four epithelial genes from **k** are shown. All beads of epithelial type are positioned, colored by normalized and log1p-transformed gene expression level. Scale bars: 500 µm.

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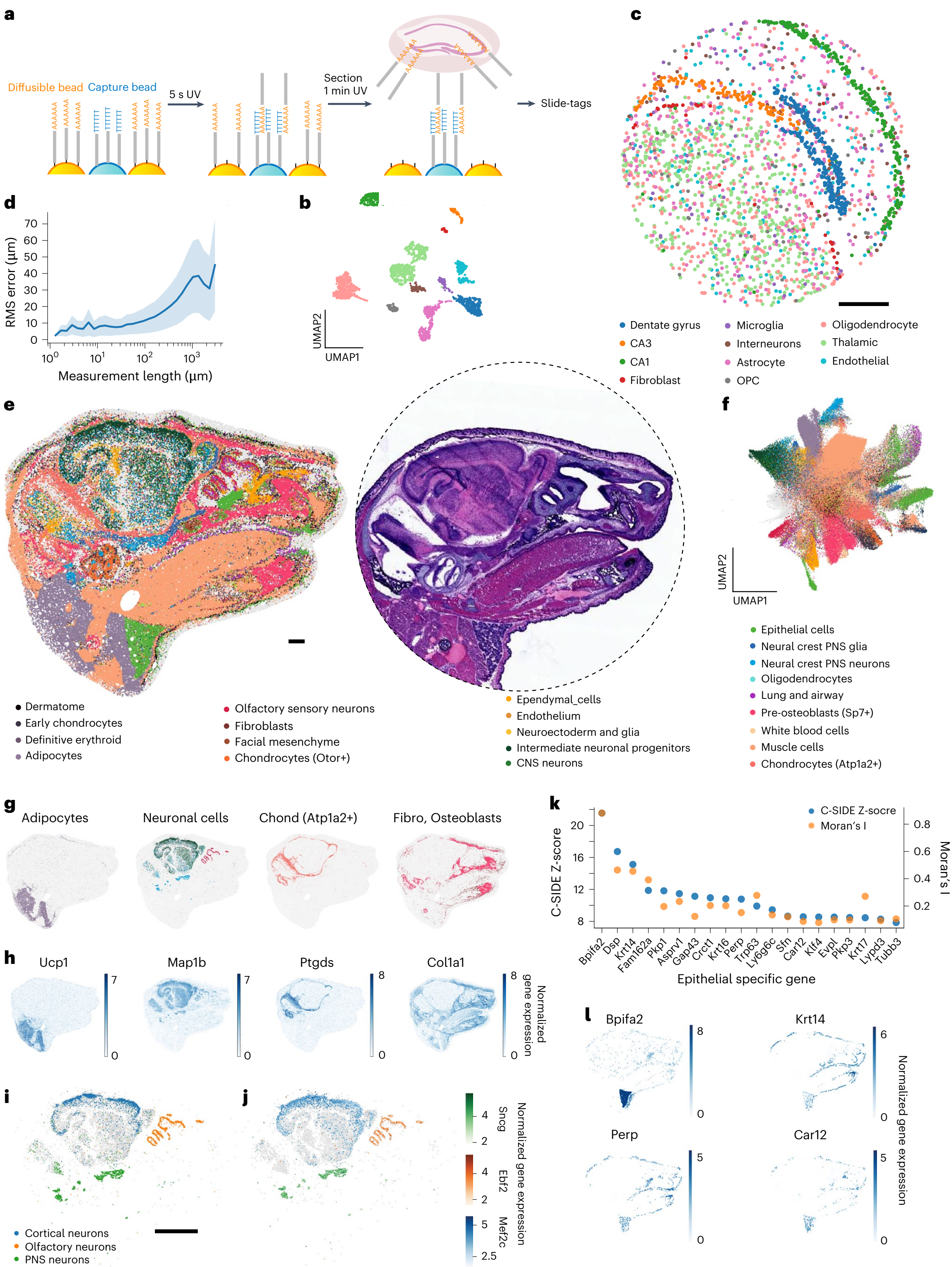

muscle cells, in a similar manner, and found 277 region-specific muscle genes (Supplementary Fig. 13 and Supplementary Table 1). These analyses demonstrate our reconstruction method can discover spatially relevant genes and revealing the complexity of gene expression patterns.

Discussion

Our work illustrates that a molecular diffusion-based computational reconstruction approach enables high-fidelity imaging-free spatial transcriptomics. We showed that UMAP can reconstruct a 2D barcode array from diffusion interactions by learning the intrinsic 2D manifold within high-dimensional data. The rationale of using a dimensionality reduction method for this reconstruction task is that the bead barcode diffusion process generates information in a high-dimensional space in which physically closer bead barcodes are also more similar. Thus, to reconstruct the physical locations, a method should preserve the high-dimensional distances while reducing it to a low-dimensional space, which matches the aim of dimensionality reduction methods. Nonlinear dimensionality reduction methods may be preferred due to the nonlinear nature of the diffusion process. Unlike t-distributed stochastic neighbor embedding (t-SNE), which primarily conserves local distances, UMAP is adept at preserving a broader spectrum of distances and outperforms other methods in accuracy and speed[23].

We systematically assessed reconstruction errors. Although the median absolute error is 25 µm, it is crucial to contextualize its impact on experiments using reconstructed data. Reconstruction introduces slight nonrigid deformations; however, these deformations are smooth, preserving local information. Thus, although a pixel may be 25 µm from its ground truth location, nearby pixels experience consistent, concordant displacements due to the smooth nature of the deformation (Extended Data Fig. 2b and Extended Data Fig. 3b,e). These deformations are comparable in magnitude and nature to tissue distortions commonly observed during processing steps like freezing, fixation and sectioning[24,25]. Local rigid registration further reduces displacements in Slide-seq and Slide-tags (Supplementary Fig. 14)

The registration process contributes to absolute error, making relative error in distance measurements a more accurate metric. The relative error in length measurements is particularly relevant because most spatial analyses depend on distance measurements, such as quantifying intercellular distances, defining cellular neighborhoods and identifying spatially varying gene expression. Here, empirical analyses of biological structures, such as the mouse hippocampus CA1 region, confirm that locally concordant errors have negligible effects on measurements of local structures or cellular neighborhoods. Though minor uncertainties may arise in absolute registration across sections, they are unlikely to affect overall conclusions.

Higher accuracy in reconstruction could potentially be attained by solving it as an assignment problem with predefined locations. Lastly, although reconstruction decouples the scale of spatial transcriptomics from imaging, the sequencing cost of reconstruction scale linearly with the array area. However, costs remain modest (Supplementary Table 2) and are expected to decrease with advances in sequencing technology.

This work demonstrates that computational reconstruction enables large-scale, high-throughput spatial transcriptomics. An elegant aspect of molecular biology tools supporting modern day genomics (for example, RNA-seq, epigenomics) has been the ability to distribute through open-source protocols and readily available enzymes. Our approach transforms spatial transcriptomics into a molecular biology tool, as opposed to one that requires specialized equipment, allowing for widespread accessibility for the scientific community. Although demonstrated with Slide-seq and Slide-tags, the experimental and computational approach may be highly generalizable to array-based capture technologies and even three-dimensional contexts. Lastly, decoupling from microscopy will enable us to perform spatial genomics at scales not limited by microscopy, to reach the dimensions of entire human organs.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41587-025-02612-0.

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Methods

Sample information and processing

Mice were group-housed under a 12 h/12 h light/dark schedule, with food and water available ad libitum. Mouse brain samples were obtained following guidelines in accordance with the US National Institutes of Health Guide for the Care and Use of Laboratory Animals under protocol number 0120-09-16 and approved by the Broad Institutional Animal Care and Use Committee. Wild-type C57BL/6 mice were anesthetized with isoflurane in a gas chamber flowing 3% isoflurane for 1 min. Blood was cleared using transcardial perfusion with a chilled pH 7.4 HEPES buffer (110 mM NaCl, 10 mM HEPES, 25 mM glucose, 75 mM sucrose, 7.5 mM $MgCl_2$ and 2.5 mM KCl). Brain was removed, frozen in liquid nitrogen vapor for 3 min and stored at −80 °C.

The C57BL/6 mouse embryo at P1 (MF-104-P1-BL) was purchased from Zyagen and stored at −80 °C before use.

Histological processing

A 12-µm-thick section from the frozen P1 mouse sample was mounted onto a glass plus slide. The Leica ST5010 Autostainer XL (Leica Biosystems) was used for hematoxylin and eosin staining. Sections were immersed in xylene, sequentially processed through 100% and 95% ethanol series and then stained with hematoxylin. Eosin staining was applied and the section was again processed through 100% and 95% ethanol series, xylene, dehydrated and covered using the Leica CV5030 Fully Automated Glass Coverslipper. The slide was imaged with Leica Aperio VERSA Brightfield, Fluorescence & FISH Digital Pathology Scanner.

Barcoded beads and array production

Bead barcodes were synthesized as in Slide-tags (ref. 20). Sequences of beads used in reconstruction for Slide-seq and Slide-tags were (1) 5′-TTTTCTACACGACGCTCTTCCGATCTJJJJJJJJTCTTCAGCGTTCC CGAGAJJJJJJJNNNNNNNNVVT30-3′ for capture beads, (2) 5′-TTT-pc-GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTJJJJJJJJTCTTCAG CGTTCCCGAGAJJJJJJJNNNNNNNNVVA30-3′ for diffusible beads (pc, photocleavable linker) and (3) 5′-PEG-pc-TTTCTACACGACGCTCTTC CGATCTJJJJJJJJTCTTCAGCGTTCCCGAGAJJJJJJJNNNNNNNNVVT30-3′ (PEG, polyethylene glycol) for capture beads in the reconstruction of P1 mouse/linker.

In Slide-seq experiments, capture beads and diffusible beads were mixed at the ratio of 3:1. In Slide-tags experiments, capture beads and diffusible beads were mixed at the ratio of 1:3. Arrays were prepared according to Slide-seqV2 (ref. 18). Beads were pelleted and resuspended in water with 10% DMSO at a concentration between 20,000 and 50,000 beads per microliter. Then 10 µl bead solution pipetted into each 3 mm well on the gasket. For a 1.2 cm array, 200 µl bead solution was used. The coverslip gasket filled with beads was centrifuged at 850 *g* for at least 30 min at 40 °C until the surface was dry, for the beads to stick to glass coverslips sprayed with Gorilla Glue and Plastic Dip. Excess beads were washed off to form a monolayer bead array. Pucks were sequenced using a sequencing-by-ligation approach with a monobase-encoding strategy[18].

Reconstruction with Slide-seq procedure

For reconstruction with Slide-seq, the protocol of Slide-seqV2 (ref. 18) was performed until the step of reverse transcription (RT). First, 10 µm frozen tissue sections were put on arrays and immersed in 200 µl hybridization buffer (6× SSC with 2 U μl^{-1} Lucigen NxGen RNase inhibitor) for 30 min at room temperature to allow for binding of RNA to the oligonucleotides on the beads. Then, arrays were incubated in RT solution (115 µl water, 40 µl Maxima 5× RT buffer (Thermo Fisher, EP0751), 20 µl of 10 mM dNTPs (NEB, N0477L), 5 µl RNase inhibitor (Lucigen, 30281), 10 µl of 50 µM template switch oligonucleotide (Qiagen, 339414YCO0076714): AAGCAGTGGTATCAACGCAGAGTGA ATrG+GrG and 10 µl Maxima H Minus reverse transcriptase (Thermo Fisher, EP0751)) for 1.5 h at 52 °C. After RT, the bead array was put on a glass slide in a 10 µl diffusion buffer (2x SSC, 20% formamide). The bead array was then placed under an ultraviolet (365 nm) light source (0.42 mW mm^{-2}, Thorlabs, M365LP1-C5, Thorlabs, LEDD1B) for 2 min (for Slide-seq on a mouse hippocampus section) or 5 s (for Slide-seq on a P1 mouse sample, as the capture beads were also photocleavable, requiring a shorter cleavage time). Then the arrays were incubated at room temperature for 10 min to diffuse. After dipping the bead array into a 1 mL diffusion buffer to wash out free oligonucleotides, the bead array was put into a 1.5 ml centrifuge tube of 200 µl extension buffer (1x NEBuffer 2, 1 mM dNTP, 25 units Klenow exo- (NEB M0212L)). The bead array was incubated at 37 °C for 1 h. Then Slide-seqV2 was continued from adding the tissue clearing buffer to cDNA PCR on the dissociated beads. A total of 200 µl tissue digestion buffer (200 mM Tris-Cl pH 8, 400 mM NaCl, 4% SDS, 10 mM EDTA and 32 U ml^{-1} proteinase K (NEB, P8107S)) was added directly to the Klenow extension solution, and the mixture was incubated at 37 °C for 30 min. Beads were pipetted up and down to detach from the surface. Then, 200 µl wash buffer (10 mM Tris pH 8.0, 1 mM EDTA and 0.01% Tween-20) was added to the 400 µl tissue clearing and RT solution mix, and the tube was centrifuged for 3 min at 3,000 *g*. The supernatant was then removed from the bead pellet, and the beads were resuspended in 200 µl wash buffer and centrifuged again. This was repeated a total of three times. The supernatant was then removed from the bead pellet, and the bead pellet was resuspended in 200 µl of 0.1 N NaOH and incubated for 5 min at room temperature. Then beads were resuspended in 200 µl of H_2O. Second-strand synthesis was then performed on the beads by incubating the pellet in 200 µl of second-strand synthesis mix (133 µl water, 40 µl Maxima 5× RT buffer, 20 µl of 10 mM dNTPs, 2 µl of 1 mM dN-SMRT oligonucleotide and 5 µl Klenow enzyme (NEB, M0210)) at 37 °C for 1 h. After second-strand synthesis, 200 µl wash buffer was added, and the beads were centrifuged for 3 min at 3,000 *g*. The supernatant was then removed from the bead pellet, and the beads were resuspended in 200 µl wash buffer and centrifuged again. This was repeated a total of three times. Water (200 µl) was added to the bead pellet, and the beads were moved into a 200-µl PCR strip tube, pelleted in a minifuge and resuspended in 200 µl water. The beads were then pelleted and resuspended in library PCR mix (22 µl water, 25 µl of Terra Direct PCR mix buffer (Takara Biosciences, 639270), 1 µl Terra polymerase (Takara Biosciences, 639270), 1 µl of 100 µM TruSeq PCR handle primer (IDT): CTACACGACGCTCTTCC GATCT and 1 µl of 100 µM SMART PCR primer (IDT)): AAGCAGTGGTATCAA CGCAGAGT, and PCR was performed according to the following program: 95 °C for 3 min; four cycles of 98 °C for 20 s, 65 °C for 45 s and 72 °C for 3 min; nine cycles of 98 °C for 20 s, 67 °C for 20 s and 72 °C for 3 min; 72 °C for 5 min; hold at 4 °C. After cDNA PCR, the beads were spun down with 2 min 3,000 RCF. Then supernatant was collected for cDNA purification, and beads were resuspended in PCR mix again for the reconstruction library. The PCR mix included 1x KAPA (Roche KK2612), 100 nM P5-Truseq Read 1 primer (5′-AATGATACGGCGACC ACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT-3′) and 100 nM P7-Truseq Read 2 primer (5′-CAAGCAGAAGACGGCAT ACGAGATNNNNNNNNGTGACTGGAGTTCAGACGTGTGCTCTTCCGA TCT-3′).

The reconstruction library PCR followed the program 3 min at 95 °C, 13 cycles of (20 s at 98 °C, 15 s at 65 °C, 15 s at 72 °C) and 1 min at 72 °C. After the PCR program, supernatant was collected after centrifuge and purified with 1.5x SPRI cleanup (Beckman Coulter, A63881), generating reconstruction library ready for sequencing. Meanwhile, the mRNA library was prepared following Nextera tagmentation[18]. Sequences used are shown in Supplementary Fig. 15.

Reconstruction with Slide-tags procedure

For reconstruction with Slide-tags, the diffusion and extension steps were performed before doing Slide-tags. Similar to the process in reconstruction with Slide-seq, the bead array was emerged with diffusion buffer (2x SSC, 20% formamide) and exposed to ultraviolet (365 nm) light source (0.42 mW mm^{-2}, Thorlabs, M365LP1-C5, Thorlabs, LEDD1B) for 5 s, incubated at room temperature for 10 min, dipped in 1 ml diffusion buffer, put in 200 µl extension buffer (1x NEBuffer 2, 1 mM dNTP, 25 units Klenow exo- (NEB M0212L)) as above and incubated at 37 °C for 1 h. After extension, the bead array was dried and used for Slide-tags protocol until nuclei extraction. Fresh frozen tissues were cryosectioned to 20 µm and then placed onto the array. The array was placed onto the glass slide in 6–10 µl dissociation buffer (82 mM Na_2SO_4, 30 mM K_2SO_4, 10 mM glucose, 10 mM HEPES, 5 mM $MgCl_2$) and exposed to ultraviolet (365 nm) light source (0.42 mW mm^{-2}, Thorlabs, M365LP1-C5, Thorlabs, LEDD1B) for 30 s. After photocleavage, the puck was incubated for 7.5 min for the oligos to tag nuclei. Nuclei were exacted and loaded into the 10x Genomics Chromium controller using the Chromium Next GEM Single Cell 3′ Kit v3.1 (10x Genomics, PN-1000268). snRNA-seq library was prepared following Slide-tags protocol[20]. After nuclei isolation from the array, beads were dissociated, washed with 200 µl wash buffer (10 mM Tris pH 8.0, 1 mM EDTA and 0.01% Tween-20) and resuspended in 200 µl PCR mix. The preparation of reconstruction is the same as in reconstruction with Slide-seq. The PCR mix included 1x KAPA (Roche KK2612), 100 nM P5-Truseq Read 1 primer (5′-AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT-3′) and 100 nM P7-Truseq Read 2 primer (5′-CAAGCAGAAGACGGCATACGAGATNNNNNNNNGGTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT-3′).

The reconstruction library PCR followed the program 3 min at 95 °C, 13 cycles of (20 s at 98 °C, 15 s at 65 °C, 15 s at 72 °C), and 1 min at 72 °C. After the PCR program, supernatant was collected after centrifuge and purified with 1.5x SPRI cleanup (Beckman Coulter, A63881), generating reconstruction library ready for sequencing. Sequences used are shown in Supplementary Fig. 16.

Sequencing

The gene expression libraries and reconstruction libraries were sequenced on the Illumina Nextseq1000 instrument using P2 100 cycle kit (Illumina, 20046811). For the P1 mouse sample, the gene expression library was sequenced on an Illumina NovaSeq instrument using the S Prime platform.

Diffusion matrix simulation

Diffusible and capture beads were simulated to locate uniformly in a circle with each bead's color determined by its location in the image of pattern 'H' or in a 2D color gradient. The diffusion of a fixed number of barcodes from each diffusible bead was assumed to follow a Gaussian distribution. Y_{ij}, the number of diffusible bead barcode j captured by capture bead i, follows a binomial distribution with the probability determined by the distance between beads:

$$Y_{ij} \sim Binomial(UMI, p_{ij})$$

$$p_{ij} = C \exp\left(-\frac{{d_{ij}}^2}{2\sigma^2}\right)$$

where d_{ij} is the Euclidean distance between the diffusible and capture beads, σ is the standard deviation in the Gaussian distribution and C is for normalization. The diffusion-based pairwise count matrix was then generated for reconstruction.

For the simulated diffusion matrix, 1,000 capture beads and 1,000 diffusible beads were located uniformly with a circle of diameter as 3,000 µm. Diffusion distance σ was 300 µm and UMI was 300.

Sequencing data processing and diffusion matrix generation

In the processing of FASTQ files from the reconstruction library sequencing, reads were filtered out if their constant sequences had Hamming distances greater than 3 when compared to the universal primer sequence. From remaining reads, capture bead barcode, capture bead UMI, diffusible bead barcode and diffusible bead UMI were abstracted. To determine the read threshold for reliable bead barcodes, rank plots were generated for both capture and diffusible bead barcodes. Barcodes above the read threshold were collapsed with a Hamming distance of 1, resulting in a whitelist of barcodes. Barcodes with reads below the threshold were matched to the whitelist with a Hamming distance of 1. Paired reads with both capture bead barcode and diffusible bead barcode in the whitelist were compiled along with the UMI information. With the whitelist, we plotted the distribution of total interaction per bead and the distribution of unique connected beads per bead for quality control.

Diffusion matrices in sparse matrix data structure were generated by taking capture bead barcodes as rows and diffusible bead barcodes as columns (or diffusible bead barcodes as rows and capture bead barcodes as columns for reconstruction with Slide-tags). The element values in these matrices corresponded to the count of unique UMIs associated with each barcode pair.

Computational reconstruction with UMAP

Diffusion matrices were used as input for UMAP[17] to reduce to a 2D space. Coordinates in the 2D space were directly used as reconstructed locations. UMAP parameters that we tuned are larger n_neighbors and larger min_dist for uniform distribution of beads, larger n_epochs for converging, and also cosine metric that can better represent the high-dimensional distance from diffusion matrix. With experimental data, we also found a log1p transformation of the diffusion matrix improved reconstruction accuracy.

For simulated diffusion matrix, we used UMAP with the following parameters: cosine metric, n_neighbores=25, min_dist=0.99, n_epochs=10000, learning_rate=1.

For reconstruction with Slide-seq or Slide-tags, we used UMAP to embed the log1p-transformed diffusion matrix with the following parameters: cosine metric, n_neighbores=25, min_dist=0.99, n_epochs=50000, learning_rate=1. The UMAP computation was expedited with 24 parallel threads.

For reconstruction with P1 mouse, we used UMAP to embed the log1p-transformed diffusion matrix with the following parameters: cosine metric, n_neighbores=45, min_dist=0.4, n_epochs=10000, learning_rate=1. The UMAP computation was expedited with 24 parallel threads. We used a smaller min_dist as: the diffusion distance (σ) was the same whereas the whole size increased, which means the relative connectivity decreased and the minimum distance between beads decreased.

Comparing reconstruction with other dimensionality reduction methods in simulation

The same simulated diffusion matrix was used for testing and comparing other dimensionality reduction methods.

PCA was performed with n_components=2 and other default parameters; MDS was performed with n_components=2 and other default parameters; Isomap was performed with n_components=2, n_neighbors=100, max_iter=100000; tSNE was performed with n_components=2, random_state=0, perplexity=50.

Evaluation of reconstruction results

To estimate the absolute errors, reconstructed locations were registered to ground truth (from simulation or in situ sequencing) using procrustes analysis[26], which applies only rigid transformation on the reconstructed locations. The reconstruction error of each bead was calculated directly by comparing the locations in ground truth and in transformed reconstruction results. The absolute errors were presented through spatial distributions, displacement vectors and distribution histograms. In the displacement vectors, only vectors with a length of less than 282.8 μm were included. In distribution histograms, errors were displayed within the indicated ranges.

To estimate the measurement length errors, pairwise distances between capture beads were calculated in both reconstruction and ground truth, with the difference between them as the measurement length errors. The measurement lengths were categorized into bins with a width of 30 μm, and the errors within each bin were averaged using the root mean square (RMS). The standard deviation for the measurement length errors within each bin was also determined. To obtain relative errors, the RMS errors were divided by the corresponding measurement lengths.

For quality control of UMAP reconstruction without ground truth, we checked the UMAP output result with uniformity and circularity and compared them with the parameters of a uniformly distributed array with the same number of beads.

Effect of UMAP parameters

We tested how UMAP parameters and normalization might affect reconstruction results with the reconstruction with Slide-seq data. The effects of UMAP n_epochs, metric ('euclidean' or 'cosine'), and log1p transformation were tested using UMAP in cuML[27] package with GPU (v.24.02.00), with other parameters kept as: n_neighbors=25 and min_dist=0.99. Results were visualized with colors determined by ground truth locations and absolute errors were calculated based on ground truth.

The effects of UMAP n_neighbors and min_dist were tested using UMAP in cuML package with GPU, with other variables as: metric = 'cosine' and n_epochs=50000. Results were visualized with colors determined by ground truth locations and absolute errors were calculated based on ground truth.

Analysis of diffusion in ground truth

The diffusion distribution of a capture bead barcode was represented by the position of its associated diffusible bead barcodes, which were color-coded according to the conjugation UMI count. The diffusion distributions of capture beads with high total UMI counts (first 3,000 beads ranked by UMI counts) along the x axis were fitted using kernel density estimation (KDE) and then averaged to derive the ensemble KDE diffusion distribution. The empirical FWHM of the diffusion distribution was calculated based on the ensemble KDE.

Slide-seq gene expression data analysis

Gene expression results were processed using the Slide-seq pipeline without bead barcode matching. We visualized the gene expression data of beads using UMAP with top 40 principal components and nearest neighbors equal to 10. Then we applied RCTD[19] to decompose the cell type associated with each capture bead barcode. The single-cell reference dataset used for comparison was the same as the one for the mouse hippocampus mentioned in the RCTD study[19].

CA1 width analysis

We profiled the spatial distribution of Atp2b1in both reconstruction and ground truth. A line perpendicular to the expression pattern of Atp2b1 in the CA1 region was drawn to characterize the expression density of Atp2b1 along this line. The widths of the CA1 region were then determined by the FWHM of the Atp2b1 expression distribution. This analysis of the CA1 width was conducted across three biological replicates.

Neighborhood enrichment analysis

The neighborhood enrichment analysis was performed using the squidpy[28] package in python (v.1.2.2). Cell types were defined by RCTD, and locations were from reconstruction and ground truth respectively. Results from reconstruction and ground truth were presented in the same scale. Pearson correlation[26] was calculated for the z-score of neighborhood enrichment from reconstruction and ground truth, with a 'two-sided' alternative hypothesis.

Matched bead and UMI per bead comparison

With the same Slide-seq library, we matched each read to a whitelist of barcodes from either in situ sequencing or reconstruction. Among all the beads matched with some gene expression reads, beads with less than 20 UMI matched were filtered for downstream analysis. The number of matched beads were calculated for in situ sequencing and reconstruction respectively. Also, the distribution of UMI per bead for in situ sequencing or reconstruction were shown with violin plot.

Slide-tags gene expression data analysis and nuclei positioning

In reconstruction with Slide-tags, single-nucleus gene expression data and spatial barcode data were analyzed with Cell Ranger[29] v.6.1.2 and Cell Bender[30] v.0.2.0 according to the Slide-tags pipeline. Nuclei positioning was performed using bead locations from both the reconstructed data and the ground truth respectively.

Evaluation of nuclei positioning error

To evaluate the error in nuclei positioning, we first transformed nuclei locations from reconstruction according to the registration between bead locations in reconstruction and ground truth. We compared nuclei locations from the reconstruction and the ground truth directly to calculate the absolute error. For the analysis of the measurement length error, we examined the pairwise distances between nuclei, employing the same methodology used for the measurement length error of beads.

Segmentation of P1 mouse bead array

A 1.2 cm circular bead array was used to profile the spatial transcriptomics of the P1 mouse section although the tissue covered only a portion of the bead array. To differentiate between the tissue-covered and uncovered regions of the bead array, segmentation was performed based on the UMI count per bead. KDE was used to estimate the UMI count density across the array, and a threshold for UMI counts was established. Only the beads covered by tissue were retained for further analysis, to save computational memory.

P1 mouse gene expression analysis

After sequencing, the gene expression results were processed using the Slide-seq pipeline. The gene expression data of the beads were visualized using UMAP, with top 40 principal components and the number of nearest neighbors equal to 10. Then we applied RCTD to decompose the cell type associated with each capture bead barcode with a reference from P1 mouse single-cell data[31]. In the reference, we excluded cell types that were not present in the tissue section being profiled.

Neuronal cell types, including CNS neurons, neural crest and PNS neurons, olfactory sensory neurons, and intermediate neuronal progenitors, were isolated and analyzed again with UMAP embedding and unsupervised clustering. Highly variable genes were found for each subcluster. Olfactory epithelium enriched genes were listed by calculating the ratio of the mean expression level in the olfactory epithelium region to that in the entire section.

To identify genes with spatial differential expression, we performed nonparametric C-SIDE[22] analysis focusing on epithelial and muscle cells. Furthermore, we calculated Moran's I statistics[28] to verify the patterned gene expression detected through C-SIDE.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data is available at the Broad Single Cell Portal: https://singlecell.broadinstitute.org/single_cell/study/SCP2577. Raw sequencing data is available on SRA: https://www.ncbi.nlm.nih.gov/sra/PRJNA1221542 (ref. 32).

Code availability

All code is available at http://github.com/Chenlei-Hu/Slide_recon.git (ref. 33).

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Author contributions

C.H. and F.C. conceived the study. C.H. developed the protocol and performed experiments with help from G.J.M., V.S. and J.A.W.; C.H. and M.B. wrote the reconstruction script. C.H. and M.B. performed analyses. V.K. synthesized the barcoded beads. C.H. and F.C. wrote the paper, with contributions from all authors.

Competing interests

F.C. is an academic founder of Curio Biosciences and Doppler Biosciences and scientific advisor for Amber Bio. F.C.'s interests were reviewed and managed by the Broad Institute in accordance with their conflict-of-interest policies. C.H., M.B. and F.C. are listed as inventors on a patent application related to the work. E.Z.M. is an academic founder of Curio Biosciences. The remaining authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41587-025-02612-0.

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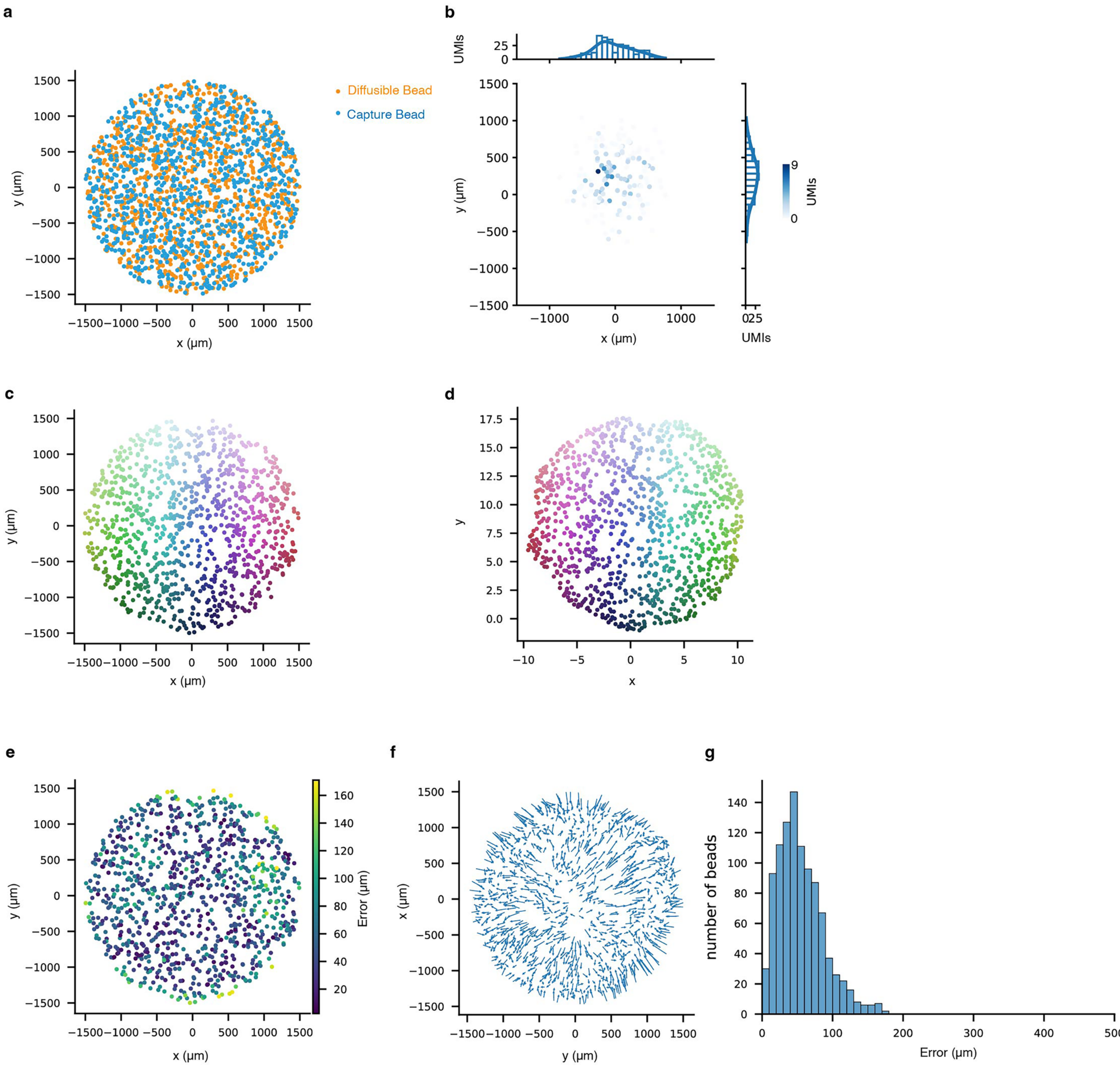
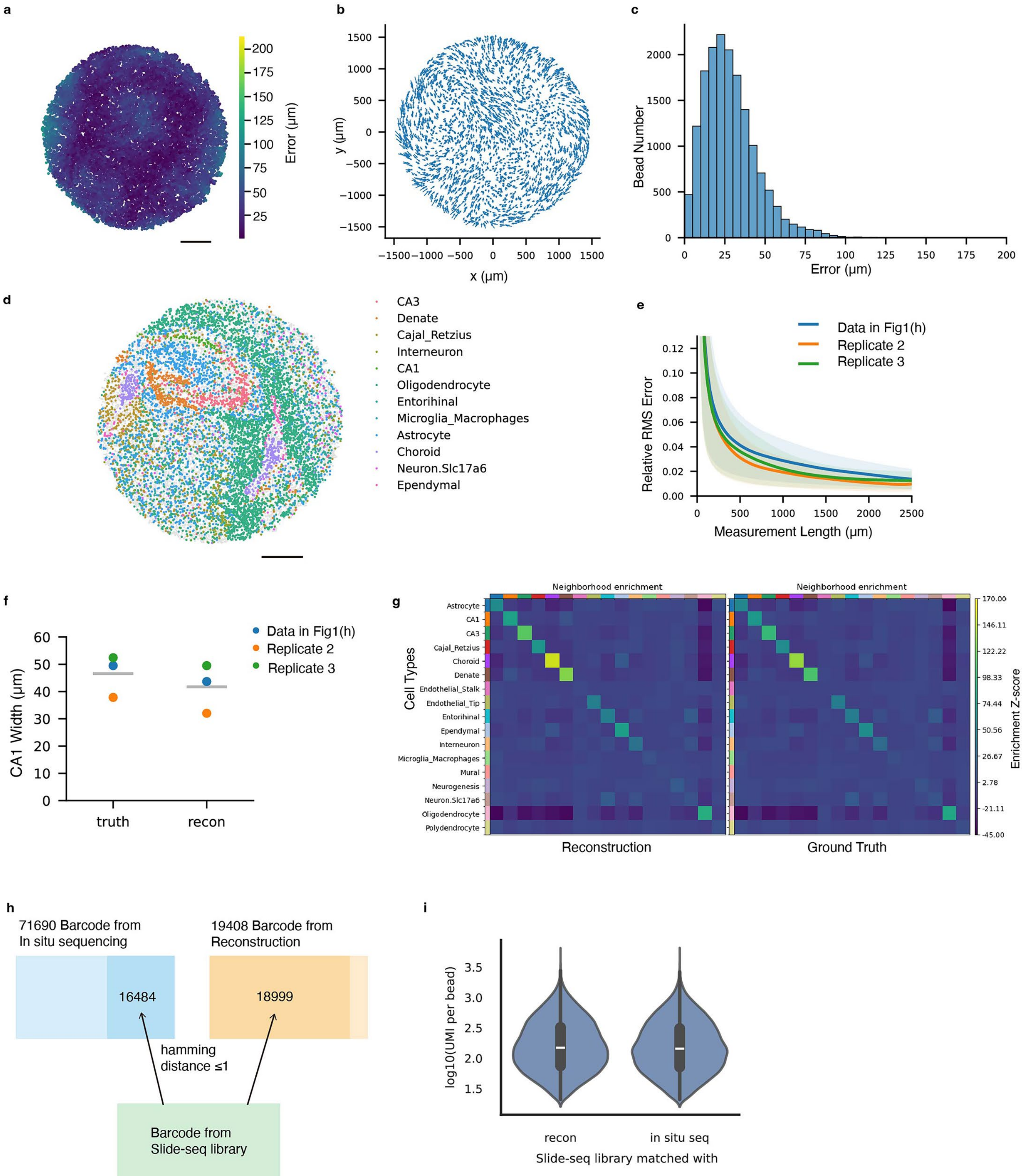
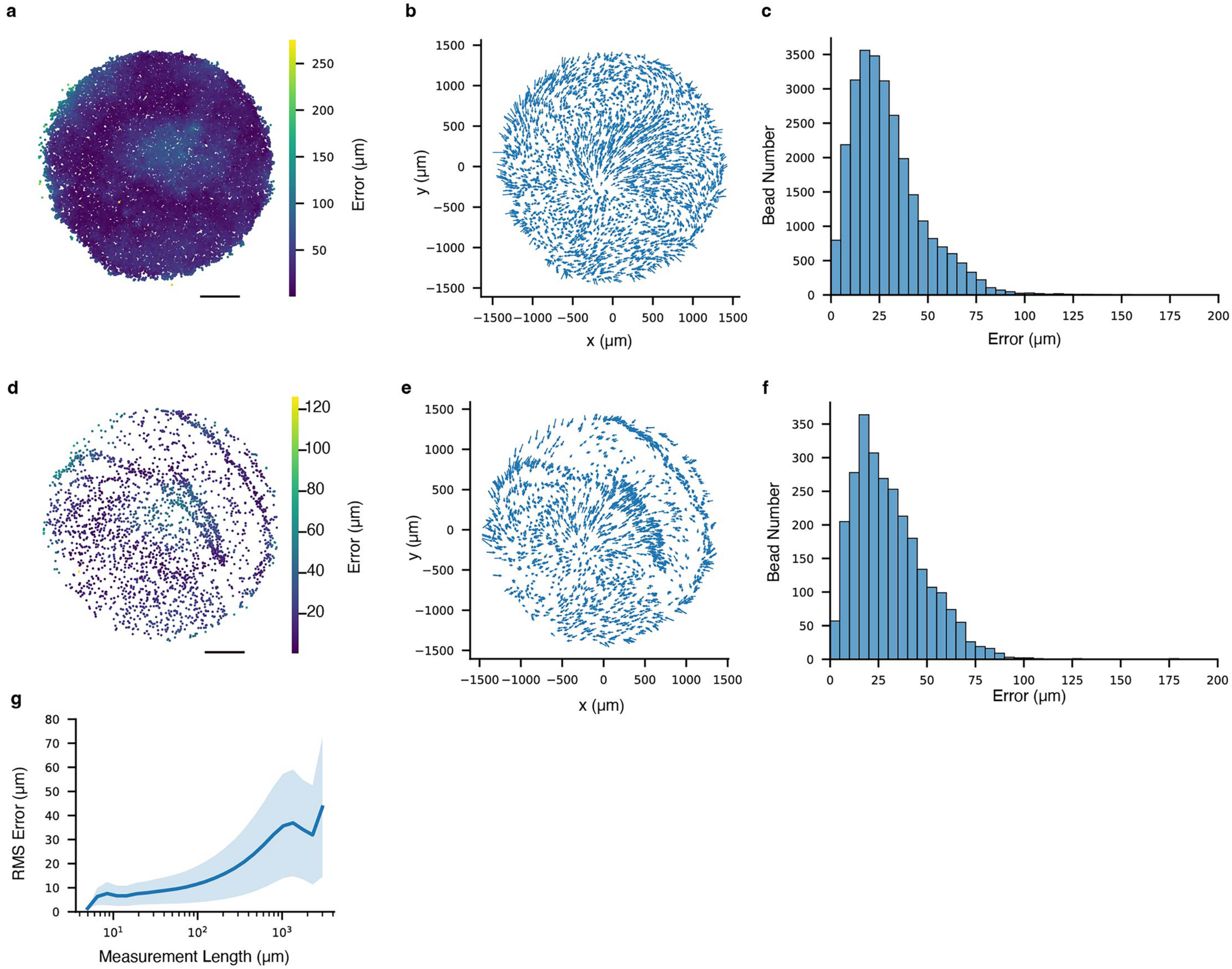


Extended Data Fig. 1 | Simulation of diffusion and reconstruction with UMAP. **a**, Simulated locations of capture beads and diffusible beads in a 3 mm circle. **b**, Simulated diffusion pattern of a capture bead on its associated diffusible beads, colored by simulated UMI counts. The distribution plots on the top and right represents the diffusion distribution on the x and y axis respectively. **c**, Simulated locations of capture beads, colored by a two dimensional color gradient depending on the locations. **d**, UMAP reconstructed locations of capture beads, colored the same as in c. **e**, Absolute error of capture beads plotted in ground truth locations. **f**, Displacement vectors of capture beads. Each arrow starts from the capture bead's ground truth location and ends at the reconstruction location. **g**, Histogram plot of capture beads' absolute error.



Extended Data Fig. 2 | See next page for caption.

Extended Data Fig. 2 | Slide-seq reconstruction metrics. a, Absolute error of capture beads plotted in ground truth locations. **b**, Displacement vectors of capture beads. Each arrow starts from the capture bead's ground truth location and ends at the reconstruction location. **c**, Histogram plot of capture beads' absolute errors. **d**, Spatial location of capture beads in ground truth, colored by decomposed cell types. **e**, Relative RMS error of measurement lengths as a function of measurement length. Data shown in Fig. 1f (blue) and two biological replicates (orange and green) are presented. Solid lines represent average values across beads and shaded areas represent one standard deviation. **f**, CA1 width measured in ground truth and reconstruction (N = 3 biological replicates). Data shown in Fig. 1f (blue) and two biological replicates (orange and green) are shown. Gray lines showed the mean width of each group. **g**, Neighborhood enrichment analysis between cell-type pairs in reconstruction (left) and ground truth (right). The enrichment scores are plotted in the same color scale, higher scores represent more enriched in the neighborhoods. **h**, Barcode matching between Slide-seq library and in situ sequencing barcode list or reconstruction barcode list. Bead barcodes with >20 UMI counts were matched with hamming distance ≤1. Blue rectangle represents total barcodes from in situ sequencing and dark blue represents barcodes matched with Slide-seq library barcodes (shown as green rectangle). Yellow rectangle represents total barcodes from reconstruction and dark yellow represents barcodes matched with Slide-seq library barcodes. **i**, Violin plot of UMI count per bead with the same Slide-seq library matched to reconstruction results and in situ sequencing results. Scale bars: 500 µm.



Extended Data Fig. 3 | Slide-tags Reconstruction metric. a, Absolute error of capture beads plotted in ground truth locations. **b**, Displacement vectors of capture beads. Each arrow starts from the capture bead's ground truth location and ends at the reconstruction location. **c**, Histogram plot of capture bead locations' absolute errors. **d**, Spatial representation of reconstruction error on each nucleus. **e**, Displacement vectors of located nuclei. Each arrow starts from the nuclei's ground truth location and ends at the reconstruction location. **f**, Histogram plot of nuclei locations' absolute errors. **f**, RMS error of measurement lengths between bead pairs as a function of measurement length. Solid lines represent average values, and shaded areas represent one standard deviation.

nature portfolio

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Reporting Summary

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	For single nuclei data collection, Cell Ranger v6.l.2, Cell Bender v0.2.0
Data analysis	For reconstruction: umap-learn v0.5.3 GPU usage: cuML v24.02.00 For result analysis: scipy vl.9.0,s quidpy vl.2.2 Packages above have been used with Python 3.11.7. Dependencies have not been listed for brevity. Code for processing spatial sequencing libraries is available on Github: https://github.com/Chenlei-Hu/Slide_recon

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- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Datasets have been deposited on the Broad Institute Single Cell Portal, under the following accession numbers: singlecell.broadinstitute.org/single_cell/study/ SCP2577. Raw sequencing data is available on SRA: https://www.ncbi.nlm.nih.gov/sra/PRJNA1221542.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	N/A
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No sample size calculation was performed. Samples sizes were chosen primarily based on experiment length, sample availablity, and sequencing costs. These sample sizes are sufficient because each sample serves as a proof-of-concept for the new technology.
Data exclusions	No data was excluded.
Replication	All attempts at replication were successful. We performed replication on reconstruction with Slide-seq datasets (3 biological replicates).
Randomization	Randomization was not applicable because the focus of this paper is the development of a new genomic technology and did not involve allocating samples/organisms/participants into experimental groups.
Blinding	Blinding was not applicable because because the focus of this paper is the development of a new genomic technology and did not involve group allocation.

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n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Mus musculus strain C57BL/6 female adult, age between 6-14 weeks. Mus musculus strain C57BL/6 Pl, age 1 day.
Wild animals	This study did not involve wild animals.
Reporting on sex	Sex was not important for this study since the tissues are used to benchmark a new protocol, which we anticipate would provide identical results regardless of sex.
Field-collected samples	No field-collected samples were used.
Ethics oversight	All procedures involving animals at the Broad Institute were conducted in accordance with the US National Institutes of Health Guide for the Care and Use of Laboratory Animals under protocol number 0120-09-16 and approved by the Broad Institutional Animal Care and Use Committee.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*